



Title – Bookstagram

“Why just read when you can create!”

Course:- Applied Research Project, CSIS- 4495_002

Team

Name	Student ID	Position
Khushi Bhatt	300394398	Team Leader
Aditi Aditi	300396361	Team Member

Introduction

Domain: The project sits in the intersection of Full-Stack Web development (MERN) and Generative AI. It specifically addresses the “Digital Publishing” and “Creator Economy” sectors.

Background: Traditionally, writing a book was a long process. However, the rise of “Visual storytelling” (as seen on Instagram/TikTok) and the democratization of AI (Google Gemini storybook, DALL-E) have shifted user expectations. Users now crave high-speed content creation and social validation (likes/followers) for their creative journeys.

Context: Bookstagram is built to bridge the gap between “Social Media” and “E-Book Authoring”. It provides a workspace where a MERN stack serves as the robust infrastructure, while AI acts as a creative partner to generate outlines, chapters, and visual “story threads”.

Problem: In today’s digital world, traditional e-books are often too long and intimidating for a generation accustomed to quick, scrolling content, leading to a significant drop in attention spans and unfinished stories.

Solution: Bookstagram solves this by turning the writing process into a social experience; it uses AI to break down long books into manageable “story threads” and engaging chapters, helping users rebuild their focus while receiving real-time encouragement from a community.

Current Research & Knowledge Gaps: Existing sources highlight that 76% of marketers use AI for basic content and publishing industry is rapidly adopting AI for cover design and editing. Moreover, there is a “Social-Authoring Gap” which shows that current tools focus only on production (AI writers) but only few provide integrated social platforms where creation process itself is the content.

Hypotheses: Integrating an AI e book generator will increase project completion rates for first-time authors by over 50%. The “Instagram-style” visual story feature will drive higher user retention than traditional text-only reading apps.

Assumptions: Current users are comfortable using Google Gemini AI as a “co-author” rather than a replacement. The Gen-Z demographic prefers “Markdown-based” clean UI over complex word processors.

Potential Benefits: For creator; it will turn difficult hobby into a fun and social experience. For readers, it offers a feed of fresh Ai stories that are more visually appealing and concise than traditional novels.

Proposed Research Project

Research Design and Objectives:-

The research follows an Experimental and Developmental Design. The goal is to build a functional prototype of “Social-Authoring” Platform to test if AI-assisted tools can mitigate “blank page syndrome” while increasing user engagement through social features.

The objective of this project is to gamify the writing process. We can achieve that by implementing social metrics (likes, follows) to provide immediate validation, addressing the lack of motivation in traditional long-form writing. Following that, we will enable multimodal storytelling by introducing a “Story thread” feature for short-form, visual content, catering to modern attention spans.

Methods and Justification:-

We will apply an Agile Software development life cycle (SDLC). This allows for iterative testing of the AI features.

Past studies on “Human-AI Co-Creativity” show that users feel more empowered when AI acts as “collaborative partner” rather than a replacement. The MERN stack is chosen because its unified JavaScript language allows for rapid prototyping and seamless data flow between the AI-driven backend and the reactive frontend.

Data Collection Methods:

The data will be collected automatically from how users interact with the Bookstagram app and saved in the MongoDB database. For each user, the system will record their user ID, how many stories and chapters they create, how often they use the AI, how many likes and followers they receive.

In addition, pre-test and post-test surveys will be connected to measure user’s writing confidence, motivation levels, perceived usefulness of AI assistance, and overall system usability.

Sample Size:-

The study will include about 15-20 participants between the ages of 12 and 25 who have beginners with intermediate writing experience.

Data analysis techniques:

User activity data will be analyzed using simple descriptive statistics such as averages to summarize writing progress and AI usage. Survey results will be compared before and after system usage to identify changes in motivation and confidence. Basic user feedback will be reviewed to understand overall user experience.

Technology Identification

Operating System – Windows 11

Programming language – JavaScript

Frameworks – React.js, Node.js with Express.js

Database – MongoDB

Style – Tailwind CSS

AI integration – Google Gemini AI

Storage – Google cloud storage

Expected Results and Practical Contributions:-

We expect that users who use the Bookstagram platform will be able to write and complete more chapters and stories than they did before, mainly because the AI helps them generate ideas and the social features keep them motivated. The survey results are also expected to show that users feel more confident and motivated about writing after using the system.

Users are likely to say that the AI helps them overcome the writer's block and makes writing easier and more enjoyable. Features such as likes and followers are expected to encourage users to write more regularly and stay engaged with their work.

In practical terms, this research can help show how combining AI with social features can improve digital writing tools. The platform could be useful in schools, writing communities, and content creation platforms to support beginner writers and make storytelling more interactive and fun.

Project Planning and Timeline (8 Weeks)

This document presents a detailed project schedule for the Bookstagram research and development project. The project will be completed within eight (8) weeks using the Agile Software Development Life Cycle (SDLC) methodology. The plan includes clearly defined phases, milestones, deadlines, deliverables, and team responsibilities.

Development Methodology:- Agile methodology is adopted to support iterative development, continuous testing, and regular evaluation of progress. The project is divided into weekly sprints with structured objectives and weekly review meetings.

Khushi Bhatt – Team Leader

- Overall project planning, task coordination, and sprint monitoring
- Backend development using Node.js and Express.js
- Frontend development using React.js and Tailwind CSS
- Database design and implementation using MongoDB
- Integration of Google Gemini AI for story generation features
- Implementation of system analytics and user activity logging
- Research data analysis and interpretation of results
- Final report writing and technical documentation
- GitHub repository management and version control supervision

Aditi Aditi – Team Member

- Participation in project planning and weekly sprint reviews
- Backend development using Node.js and Express.js
- Frontend development using React.js and Tailwind CSS
- Database design and data handling using MongoDB
- Implementation of social features (likes, follows, story feed)
- System testing, debugging, and quality assurance
- Design and administration of pre-test and post-test surveys
- Preparation of project presentation slides and system demonstration materials

Week 1 – Requirements & System Design

Goals:

- Finalize functional requirements (AI features, social features, user roles)
- Design system architecture (MERN + Gemini AI)
- Create wireframes for:

- Login/Register
- Story creation
- AI assistant panel
- Story feed
- Profile page

Tasks:

- Define database schema (Users, Stories, Chapters, Likes, Followers)
- API endpoint planning
- Task division between team members

Deliverables:

- Requirement specification
 - UI wireframes
 - Database design
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Week 2 – Environment Setup & Backend Foundation

Goals:

- Set up development environment
- Initialize backend project

Tasks:

- Setup GitHub repository
- Setup Node.js + Express server
- Configure MongoDB
- Implement authentication (JWT)
- User model & basic CRUD APIs

Deliverables:

- Working backend server
 - User authentication system
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Week 3 – Frontend Foundation

Goals:

- Build core UI structure

Tasks:

- Setup React project
- Integrate Tailwind CSS
- Create pages:
 - Login/Register
 - Dashboard
 - Story Editor
 - User Profile
- Connect frontend with backend APIs

Deliverables:

- Functional frontend layout
 - Auth working end-to-end
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Week 4 – AI Integration (Google Gemini)

Goals:

- Implement AI writing assistance

Tasks:

- Integrate Gemini API
- Features:
 - Story outline generation
 - Chapter generation
 - Idea suggestions
- Store AI-generated content in database

Deliverables:

- AI-assisted story creation working
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Week 5 – Social Features Implementation

Goals:

- Build “social authoring” experience

Tasks:

- Like system
- Follow system
- Story feed (Instagram-style)
- User profiles with stats

Deliverables:

- Social interaction fully functional
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Week 6 – Story Threads & UI Improvements**Goals:**

- Implement short-form visual storytelling

Tasks:

- Story threads feature
- Markdown-based editor improvements
- UI optimization for mobile responsiveness
- Image upload support (Google Cloud storage)

Deliverables:

- Complete feature set
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Week 7 – Testing & Data Collection Setup**Goals:**

- Prepare for research study

Tasks:

- Bug fixing
- Performance testing
- Create pre-test & post-test surveys
- Implement analytics logging:
 - AI usage
 - Chapters created
 - Likes & followers
- Recruit participants (15–20 users)

Deliverables:

- Stable prototype
 - Survey ready
 - Data tracking active
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Week 8 – Evaluation, Documentation & Presentation**Goals:**

- Finalize research output

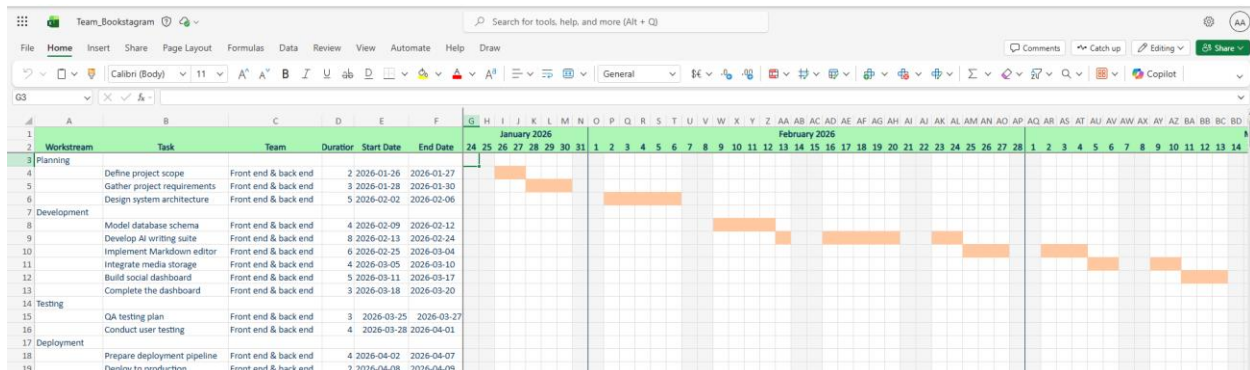
Tasks:

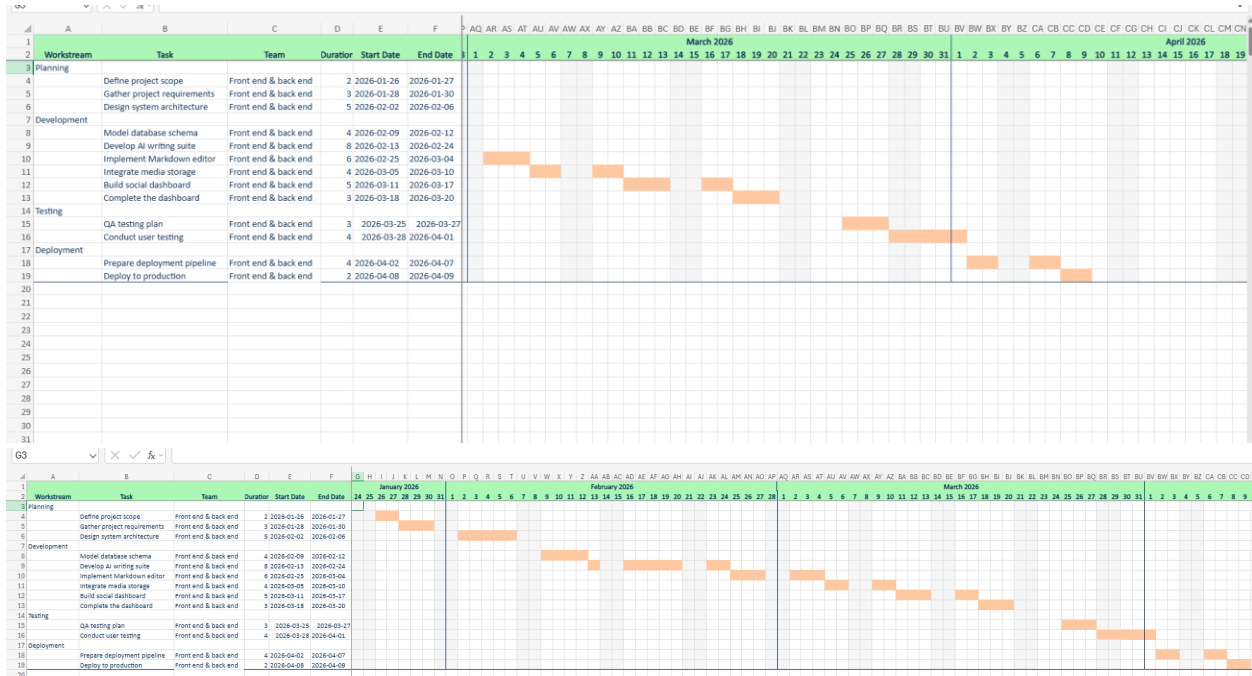
- Conduct user testing
- Collect & analyze data
- Prepare:
 - Final report
 - Screenshots
 - Architecture diagram
 - Results charts
 - Presentation slides
- Final code cleanup

Deliverables:

- Completed project
- Research findings
- Final submission

Project management chart





Project Contract

This contract confirms that both team members will contribute equally to succeeding in the project

- 1) Members will update each other regularly on findings and status of tasks being carried out via WhatsApp.
- 2) Team meetings will be carried out once a week in order to ensure that the project is on schedule and on the right track.
- 3) Both the team members will involve in research, design, and documentation and presentation.
- 4) Tasks will be divided among themselves fairly and rotated when possible.
- 5) Decisions will be taken by discussing with the other member and meetings will be hosted wherever required before taking decisions.
- 6) Members respect each other's ideas.
- 7) All relevant documents will be maintained in the GitHub Repo and will be accessible to each other.

Signatures

Name: Khushi Bhatt
Signature: Khushi
Date: 19-01-2026

Name: Aditi Aditi
Signature: Aditi
Date: 19-01-2026

AI Use Section

AI Tool Name	Version, Account Type	Specific feature for which the AI tool was used	Value Addition What value did you add over and above what AI did for you?
Chatgpt	Free version	For paraphrase	I reviewed the generated text, selected relevant parts, rewrote sections to match my project context (Bookstagram), added technical details, and ensured the content aligned with the proposal requirements and rubric.
Free Grammer checker	Free version	Grammer checker	I manually reviewed all corrections, adjusted sentence structure for clarity, and ensured the academic tone was maintained instead of directly copying the suggested changes.
Google Gemini AI	Free version	Generating ideas for project	I evaluated the generated ideas, selected useful ones, customized them to fit the project goals, and converted them into practical features for the Bookstagram platform.

Work Logs

Khushi's work logs

Date	Number of Hours	Description of work done
11/1/2026	1	Work on finding project ideas
13/1/2026	2	Work on the project ideas and market research
16/1/2026	3	Finalizing project Scope and research about core concepts
17/1/2026	2	Work on the project proposal
20/1/2026	2	Work on background research and writing.
22/1/2026	1	Work on timeline and requirement phase
24/1/2026	2	Requirement phase and planning phase
25/1/2026	1	Completing proposal document

Aditi's Work logs

Date	Number of Hours	Description of work done
11/1/2026	1	Work on brainstorming the project idea
12/1/2026	1	Search on social media like you-tube ,indeed, about the scope of market
13/1/2026	1	Collect different ideas from the job-description
16/1/2026	2.5	Finalizing the end product and start to search on the big scope
18/1/2026	1	Work on the project proposal
19/1/2026	1	Work on the project proposal and gathering information of technology
22/1/2026	1	Planning phase
24/1/2026	1	Planning phase

References

<https://www.combinedbook.com/how-ai-is-changing-the-landscape-of-book-publishing-and-copyright/>

<https://vistasocial.com/insights/generative-ai/>

<https://docs.cloud.google.com/vertex-ai/generative-ai/docs/image/overview>

<https://ai.google.dev/gemini-api/docs>

<https://www.youtube.com/watch?v=kojaR6LBdzI>